Group Project Instructions & Deliverables

Course: ISA 345

Instructor: Dr. Maliha Alam

Points:100

Group size: 4 to 5

1. FOCUS AND CONNECTION WITH COURSE GOALS

This course project builds on several of our class topics and requires you to apply your knowledge of database concepts, design, and technology to solve a specific business problem. The project has several requirements and addresses the following course goals:

1. Analyze and document the information needs of a business problem using relational database models.
2. Design a relational database to address challenging and complex requirements.
3. Code a database design based using the DDL properties of SQL.
4. Write simple test queries, advanced data retrieval, and maintenance commands using SQL.
5. Present technical reports in a professional, business manner.

The deliverable for this project is a report describing the general business setting, the underlying problem a client/user is facing, and the proposed solution.

2. PROJECT REPORT

Each group must prepare and submit a report before the deadline. In your report, you must organize and explain all the work your group did. This includes the documentation regarding the database analysis, design, implementation, and operations. <u>Remember to be professional: your report must be pleasant to read!</u> Prepare the final report with a cover page containing the project name, group members' names, date, class, and session. Before submission, proofread your report to remove typos and grammar mistakes. If you add print screens or figures to your report, make sure they are readable and good-looking (yes, good-looking!). This is a business report: your group might lose many points by submitting sloppy reports. Your report (a pdf file) must be submitted on Canvas before the deadline.

2.1 Report Structure

Your report must have the following sections, which are better explained in the following subsections:

1. Project Overview [10 points]: describe the business process, activities, and transactions.

2. Database Design and Implementation [35 points]: describe the design of the database, including any interesting decisions made. Describe how the design was implemented, including potential problems the group faced and how these problems were solved.

3. Operational SQL [45 points]: explain the queries that were created and how they will improve the underlying business process. Illustrate via a basic webpage how an end-user would interact with the proposed database.

4. Conclusions & Lessons Learned [10 points]: reflect upon the process and discuss how the proposed system addresses the underlying problem.

5. Appendices
    A. Oracle DDL statements: CREATE TABLE statements
    B. Oracle DML statements: INSERT statements
    C. HTML/PHP code concerning the developed webpage

2.2 Project Overview [10 points]

The first step in this project (actually, in any database engineering project) is to clearly identify the problem and domain your group will be working on. You can embrace your inner entrepreneurial spirit and create business processes for a hypothetical organization. Think about a company, product, or service you would like to develop that would require the use of a database. You are also allowed to use the Internet to find an "application" or process that requires the use of a database as long as the same has not been discussed in class. Examples of previous projects include: 1) a database to store data related to baseball (players, teams, results of games, …). These data were supposedly used by a sports-related website; 2) a database to store data related to a restaurant (orders, inventory, clients, employees, …); and 3) a database to store data related to a new online payment system similar to PayPal (transactions, users, …).

The underlying business processes should include many *operational transactions* that use the designed database. For example, a bank-like mobile app would allow users to transfer money and check an account balance. The process of transferring money involves at least two *database transactions* (which are *not* the same as operational transactions): 1) withdrawing money from one account and 2) depositing money into a second account. Keep in mind that operational transactions are not necessarily financial. For example, creating a new user or shipping a product are also transactions in e-commerce domains. You should create a narrative description of the processes and operational transactions being supported. Explicitly describe at least three operational transactions the designed database supports.

2.3 Database Design and Implementation [35 points]

After defining the problem and domain of interest, your next step is to design a relational database using a logical model (see Lectures 17 and 18 for more details). Besides including a graphical representation of the model in your report, you must also briefly explain (define) each entity and all of its attributes. Your model should have at least eight entities. Keep in mind that it is very unlikely that an entity will exist on its own. Instead, each entity will likely be in relationships with other entities. For each relationship in your model, write the business rules the relationship is encoding, including both minimum and maximum cardinalities.

Discuss any issues with the normalization process for each of the entities in your model. In particular, list the functional dependencies in your relational tables involving the PK that might cause violations of normal forms. Refine the original logical model so as to remove redundancies from your tables. See Lectures 19 and 20 for more details on normalization.

Once you are satisfied with your logical model, it is now time to design a physical database model. In particular, specify attribute types and domain constraints whenever required. You do not have to explain your choices of attribute types, but they must be accurate given the problem domain. Besides the logical model, you must also add a graphical representation of the physical model in your report.

Using Data Definition Language (DDL) commands, you must next implement your design using Oracle DBMS. Be sure to implement all the constraints when creating your tables, *i.e.*, to establish the appropriate business rules and relationships. Remember that the SQL code must mirror the database design. All the CREATE TABLE statements must be added to the appendix of your report (see Section 2.1). *One must be able to copy all the statements from your report, paste them into SQL Developer, and run without any errors*.

Insert enough data into your database so as to support realistic queries. You should have at least ten rows per table. The data should be as realistic and consistent as possible. In particular, *one must be able to copy all the INSERT statements from your report, paste them into SQL Developer, and run without any errors*. All the INSERT statements must be added to the appendix of your report (see Section 2.1).

2.4 Operational SQL [45 points]

The next phase of the project is to demonstrate the managerial value of the database. To this end, you have to define at least five complex queries that demonstrate how users could benefit from the information provided by the database. You have to select your queries carefully. In particular, think about queries that would not embarrass you in a demo meeting with key stockholders (such as *SELECT * FROM TABLE*). At the same time, be sure to demonstrate your SQL prowess! In total, your queries must cover the following requirements (3 points each):

1. Contain a subquery
2. Contain an inner join
3. Contain an outer join
4. Contain a GROUP BY clause without the HAVING clause
5. Contain a GROUP BY clause together with the HAVING clause

You must explain each query and the information it provides to the user, manager, or transaction being supported, *i.e.,* link the query to a business solution. Besides the above queries, you should also perform the following operations (5 points each):

6. Define relevant INDEXES
7. Define a relevant VIEW
8. Define a relevant PROCEDURE
9. Define a relevant TRIGGER

For each index, you must explain the reason why the index was created. Similarly, you have to explain the reason why the VIEW, PROCEDURE, and TRIGGER were created and what they accomplish. Your PROCEDURE must have more than one SQL statement, *e.g.*, two updates or one insert and one update, *etc*.

2.5 Conclusions & Lessons Learned [10 points]

At the end of the report, you must write a conclusion section that explains the benefits of the proposed database system. Make a case for your application: try to convince a user/client/investor of the benefits of this database and your professional services by considering the information needs of the client/problem and how your specific queries address them.

Emphasize the improved decision-making, productivity gains, or additional controls you have in place that promote data quality. Demonstrate that you have successfully achieved the original objectives you have stated for this project (or explain why this was impossible).

Document any decisions that affected your design, implementation, or operations. Discuss any problems or issues that you encountered. Describe any changes that you made from your original plan to the end product.

3. OTHER CONSIDERATIONS

Every group member must individually fill in the document called "Peer Evaluations.docx", which is available on Canvas, and upload the same on Canvas before the project deadline. A student will only receive his/her project grade after uploading this file. Peer evaluations will be used to adjust individual grades. Any free-riding behavior will result in severe point deductions.